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INSTRUCTOR REFERENCE

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MSS TRAINING PROGRAM — INSTRUCTOR OVERVIEW

Maven Smart System (MSS) — USAREUR-AF

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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MAVEN SMART SYSTEM (MSS) — USAREUR-AF

Field	Detail
Program	Maven Smart System (MSS) Training Program
Owner	USAREUR-AF Operational Data Team
Version	1.0 — March 2026
Audience	Instructors, lead instructors, training OICs, and newly onboarded training NCOs

BLUF

This document gives a qualified instructor everything needed to understand the MSS training program and begin teaching. All documents referenced below exist in this repository. Nothing needs to be built from scratch — load the environment, print the materials, and execute the plan.

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AUTHORITATIVE REFERENCES

Publication	Title	Relevance
AR 350-1	Army Training and Leader Development	Master regulation for Army training policy; governs instructor qualification requirements
TR 350-70	Army Learning Policy and Systems	TRADOC master regulation governing instructor certification, course delivery standards, and learning products
TP 350-70-3	Faculty and Staff Development Program	TRADOC pamphlet governing instructor development, certification, and sustainment training
TP 350-70-7	Army Educational Processes	TRADOC pamphlet governing instructional methodology, assessment design, and evaluation
FM 7-0	Training	Unit training management procedures; provides context for how institutional training integrates with unit training plans

NOTE

TR 350-70, TP 350-70-3, and TP 350-70-7 are published by TRADOC at adminpubs.tradoc.army.mil, not DA APD.

1. PROGRAM OVERVIEW

The MSS Training Program is a tiered, progressive curriculum that trains USAREUR-AF personnel to operate, build on, and lead with the Maven Smart System.

Training Track Architecture

```
TM-10 → TM-20 → TM-30 → TM-40 (all tracks) → TM-50 (specialist advanced only)

T3-I (Instructor Certification)     ← prereq: TM-30 + C2DAO selection
T3-F (MSC Force Multiplier)         ← prereq: TM-20 + CDR nomination
```

Level	Course	Duration	Audience	Prereq
TM-10	Maven User	1 day	All MSS users	None
TM-20	Builder	5 days	Analysts, data builders	TM-10
TM-30	Advanced Builder	5 days	Senior analysts, pipeline builders	TM-20
TM-40A	Intelligence WFF	3 days	G2/S2 staff	TM-30
TM-40B	Fires WFF	3 days	FSE, targeting teams	TM-30
TM-40C	Movement & Maneuver WFF	3 days	G3/S3 maneuver staff	TM-30
TM-40D	Sustainment WFF	3 days	G4/S4, FSB/BSB staff	TM-30
TM-40E	Protection WFF	3 days	Force protection officers, CBRN	TM-30
TM-40F	Mission Command WFF	3 days	G3/S3, battle captains, XOs	TM-30
TM-40G	ORSA	5 days	FA49, data scientists	TM-30
TM-40H	AI Engineer	5 days	AI engineers, AIP practitioners	TM-30
TM-40M	ML Engineer	5 days	ML engineers	TM-30
TM-40J	Program Manager	4 days	MSS program managers	TM-30
TM-40K	Knowledge Manager	4 days	KMs, S6 data teams	TM-30
TM-40L	Software Engineer	5 days	SWEs, Foundry developers	TM-30
TM-50G	Advanced ORSA	5 days	Senior FA49, theater ORSA	TM-40G
TM-50H	Advanced AI Engineer	5 days	Senior AI engineers	TM-40H
TM-50M	Advanced ML Engineer	5 days	Senior ML engineers	TM-40M
TM-50J	Advanced Program Manager	3 days	Senior PMs	TM-40J
TM-50K	Advanced Knowledge Manager	3 days	Senior KMs	TM-40K
TM-50L	Advanced Software Engineer	5 days	Senior SWEs, platform leads	TM-40L

Critical note: There is no TM-50A through TM-50F. TM-50 is specialist advanced only (G–O). All TM-40 WFF tracks (A–F) are terminal — graduates proceed to their operational role, not to a TM-50.

2. WHERE EVERYTHING LIVES

All documents are under `maven_training/`. The directory structure is:

```
maven_training/
├── INSTRUCTOR_OVERVIEW.md          ← this document
├── README.md                       ← program summary, track table, prereq tree
├── QUICK_START.md                  ← 1-page orientation for new students
│
├── tm/                             ← Course content (read by students and
instructors)
│   ├── TM_10_maven_user/
│   │   └── TM_10_MAVEN_USER.md
│   ├── TM_20_builder/
│   │   └── TM_20_BUILDER.md
│   ├── TM_30_advanced_builder/
│   │   └── TM_30_ADVANCED_BUILDER.md
│   ├── TM_40A_intelligence/        ← (and TM_40B through TM_40L)
│   │   ├── TM_40A_INTELLIGENCE.md
│   │   └── CONCEPTS_GUIDE_TM40A_INTELLIGENCE.md
│   ├── TM_50G_orsa_advanced/       ← (and TM_50H through TM_50L)
│   │   ├── TM_50G_ORSA_ADVANCED.md
│   │   └── CONCEPTS_GUIDE_TM50G_ORSA_ADVANCED.md
│   ├── T3_I_instructor_certification/
│   │   ├── T3_I_INSTRUCTOR_CERTIFICATION.md
│   │   └── CONCEPTS_GUIDE_T3I_INSTRUCTOR_CERTIFICATION.md
│   └── T3_F_msc_force_multiplier/
│       ├── T3_F_MSC_FORCE_MULTIPLIER.md
│       └── CONCEPTS_GUIDE_T3F_MSC_FORCE_MULTIPLIER.md
│
├── syllabi/                        ← One syllabus per course (schedule, learning
objectives)
│   ├── SYLLABUS_TM10.md
│   ├── SYLLABUS_TM20.md
│   ├── SYLLABUS_TM30.md
│   ├── SYLLABUS_TM40A.md           ← (through TM40L)
│   └── SYLLABUS_TM50G.md           ← (through TM50L)
│
├── exercises/
│   ├── README.md                   ← exercise index
│   ├── exams/                      ← all PRE and POST exams (with embedded answer
keys)
│   │   ├── EXAM_TM40A_PRE.md
│   │   ├── EXAM_TM40A_POST.md
│   │   └── ...                     ← complete for all TM-10 through TM-50 series
│   ├── EX_10_operator_basics/
│   │   ├── EXERCISE.md
│   │   └── ENVIRONMENT_SETUP.md
│   ├── EX_20_no_code_builder/
│   ├── EX_30_advanced_builder/
│   ├── EX_40A_intelligence/        ← (through EX_40L)
│   └── EX_50G_orsa/               ← (through EX_50L)
│
├── training_management/            ← Program administration documents
│   ├── INSTRUCTOR_OVERVIEW.md      ← this document
│   ├── MTP_MSS.md                  ← Mission Training Plan
│   ├── POI_MSS.md                  ← Program of Instruction
│   ├── CAD_MSS.md                  ← Course Administrative Data
│   ├── TEO_MSS.md                  ← Training and Evaluation Outline
```

```
│   ├── ENROLLMENT_SOP.md           ← Enrollment procedures
│   ├── ANNUAL_TRAINING_SCHEDULE.md ← Schedule and throughput targets
│   ├── FACULTY_DEVELOPMENT_PLAN.md ← Instructor qualification standards
│   └── lesson_plans/
│       ├── TM10/TM10_LESSON_PLANS.md
│       ├── TM20_LESSON_PLAN_OUTLINES.md
│       ├── TM30_LESSON_PLAN_OUTLINES.md
│       ├── TM40_WFF_LESSON_PLAN_OUTLINES.md     ← TM-40A through TM-40F
│       ├── TM40_SPECIALIST_LESSON_PLAN_OUTLINES.md  ← TM-40G through TM-40O
│       └── TM50_ADVANCED_LESSON_PLAN_OUTLINES.md   ← TM-50G through TM-50O
│
├── doctrine/                       ← Reference doctrine (not course-specific)
│   ├── DATA_LITERACY_technical_reference.md
│   ├── DATA_LITERACY_senior_leaders.md
│   └── GLOSSARY_data_foundry.md
│
├── standards/
│   └── NAMING_AND_GOVERNANCE_STANDARDS.md
│
└── pdf/                            ← Generated PDFs of all above documents
```

3. WHAT TO READ BEFORE YOU TEACH

Before your first course (any level):

1. **This document** — you're reading it now
2. `training_management/POI_MSS.md` — official program of instruction; defines what you are authorized to teach and how
3. `training_management/TEO_MSS.md` — training and evaluation outline; defines the Go/No-Go standards for every evaluation
4. `training_management/FACULTY_DEVELOPMENT_PLAN.md` — your own instructor qualification requirements; verify your Go is on file before teaching

Before teaching a specific course:

1. **The TM** — read the full TM for the course you are teaching (`tm/TM_XX_name/TM_XX_NAME.md`); if a Concepts Guide exists for the course, read it too
2. **The syllabus** — `syllabi/SYLLABUS_TMxx.md`; this is your day-by-day schedule and learning objectives
3. **The lesson plan** — `training_management/lesson_plans/[relevant outline]`; each block has a TLO, delivery method, student activity, and common errors table

4. **The exercise** — `exercises/EX_XX_name/EXERCISE.md` and `ENVIRONMENT_SETUP.md`; do the setup personally before Day 1, not on the morning of the exercise
5. **Both exams** — `exercises/exams/EXAM_TMxx_PRE.md` and `EXAM_TMxx_POST.md`; the answer key is embedded at the bottom of each file under "INSTRUCTOR USE ONLY" — do not distribute the full file to students

4. PRE-COURSE CHECKLIST (ALL LEVELS)

Complete these steps **no later than 5 duty days before Day 1** unless otherwise noted:

Admin

- ☐ Verify enrollment is finalized — ENROLLMENT_SOP.md defines the process; verify students have documented prereqs on file
- ☐ Confirm room and equipment (projector, whiteboard, student workstations)
- ☐ Print PRE exam (student version — the sections above the answer key divider line)
- ☐ Confirm your instructor Go evaluation is current and on file with the training NCO

Training Environment (MSS Access)

- ☐ Verify student training accounts are provisioned and accessible — have at least one student attempt login 2 days before Day 1
- ☐ Load synthetic training data for the practical exercise — see `ENVIRONMENT_SETUP.md` for the specific course; do this yourself to verify before exercise day
- ☐ For TM-50G/H/I/L: confirm Code Workspace with GPU allocation — **10 duty days minimum**; this is the most commonly missed lead time
- ☐ For TM-50L: confirm CI/CD pipeline access — **10 duty days minimum**; separate from Code Workspace access

Evaluator

- ☐ If you are not the evaluator for the Day 3/5 practical exercise, confirm your evaluator is briefed and has read the EXERCISE.md
- ☐ Evaluator must have a separate evaluator account with the permissions described in ENVIRONMENT_SETUP.md
- ☐ Print and review the TEO section for the course you are teaching — the TEO defines the Go/No-Go criteria that override the instructor's individual judgment

5. EXAM ADMINISTRATION

PRE-TEST (Day 1, Morning)

- Administer before any instruction begins
- 20 minutes, paper or digital
- Diagnostic only — score does not affect enrollment or course outcome
- Collect, review during lunch on Day 1, and use results to calibrate depth of instruction for identified gaps
- Answer key is embedded in the exam file — do not distribute the file; score manually

POST-TEST (Day 3 or Day 5, after practical exercise)

- 30 minutes, paper or digital
- Passing score varies by course (see the exam file header for the passing score)
- A failing score does not automatically disqualify a student from a Go on the practical exercise — the two assessments are independent
- Both the practical exercise evaluation form and the POST test score go into the training record

6. PRACTICAL EXERCISE EVALUATION

The practical exercise is the primary evaluation for every TM-40 and TM-50 course. Key rules:

1. **Go/No-Go is task-based.** Each task has explicit Go and No-Go criteria in EXERCISE.md. The evaluator scores each task independently; the overall course Go/No-Go is derived from the task scores per the scoring rule at the bottom of the exercise (e.g., "Go on 5 of 6 tasks = overall Go").
2. **Automatic No-Go tasks exist.** Each exercise designates 1–2 tasks as automatic No-Go tasks — a No-Go on these tasks results in an overall course No-Go regardless of other task scores. These tasks reflect the core competency the course is designed to establish.
3. **Data staleness/inject events are scripted.** Exercises with inject events (e.g., EX_40A's SIGINT feed staleness inject at T+90 min) must be executed at the prescribed time. Do not hint at the inject before it occurs.

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4. **Coaching vs. evaluation.** During the exercise, do not coach. Answer direct questions about the training environment (environment issues are not evaluation events), but do not provide procedural hints. The line: "Can you show me what you're seeing on your screen?" is observation, not coaching.
5. **Evaluation forms go to the training NCO** within 24 hours of exercise completion. Overall Go/No-Go, participant name, evaluator name, and date are required fields.

7. T:I RATIOS AND CLASS CAPS

Level	Max Students	Min Instructors	Notes
TM-10	24	1	Lab sessions: 2 instructors recommended
TM-20	18	1	Day 4–5 labs: 2 instructors required
TM-30	12	1	Day 5 PE: dedicated evaluator required
TM-40 (WFF A–F)	12	1	Day 3 PE: dedicated evaluator required; T:I ≤ 6:1 for PE
TM-40 (Specialist G–O)	10	1	Day 3 PE: dedicated evaluator; technical depth requires T:I ≤ 5:1
TM-50 (all)	8	1	Day 5 PE: dedicated evaluator; peer review component requires instructor presence throughout

If enrollment exceeds these caps, split sections. Do not exceed class maximums — the exercise environments are not designed for larger groups and evaluator attention cannot be maintained.

8. COMMON MISTAKES TO AVOID

These are the most frequently recurring errors observed in MSS course delivery. They are documented here because they are non-obvious and not covered in the TM itself.

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Mistake	What It Looks Like	Correct Approach
Running the practical exercise on the morning of Day 3/5	Environment issues discovered at 0750 on exercise day with no time to resolve	Complete ENVIRONMENT_SETUP.md setup the day before; personally verify all data loads and test data fires the alerts
Distributing the full exam file	Student receives the file with the answer key at the bottom	Print the student version only (everything above the "ANSWER KEY — INSTRUCTOR USE ONLY" divider); never share the file itself
Coaching during the practical exercise	Telling a student what button to click when they're stuck	Observe and document; redirect environment questions; do not provide procedural hints
Marking a student No-Go without evaluator form	Verbal Go/No-Go communicated; no written record	All Go/No-Go decisions require the evaluation form submitted to the training NCO; verbal is insufficient
Teaching TM-40 WFF content to students without TM-30 prereq	Student enrolled "by exception"	The prereq is a hard requirement, not a recommendation; a student without TM-30 documented cannot be evaluated to the TM-40 standard; escalate to the training NCO before Day 1
Conflating reported status and assessed status when demonstrating the COP	Instructor describes a data layer as "showing current status" without noting it is reported data from the last submission	Model the language explicitly: "This shows reported status as of [timestamp]. The commander's assessment may differ." This is a core distinction across all WFF tracks
Skipping the PRE test	"We ran long on logistics; we'll just start the instruction"	The PRE is how you know what to emphasize on Day 1; administer it first, even if you don't have time to score it until break

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9. INSTRUCTOR CERTIFICATION PROCESS

This section establishes the certification pathway for MSS instructors per TP 350-70-3, *Faculty and Staff Development Program*. For detailed instructor qualification standards, prerequisites, and evaluator criteria, see `training_management/FACULTY_DEVELOPMENT_PLAN.md`.

9-1. Certification Pathway

All MSS instructors must complete the following progression before independently delivering instruction:

Phase	Activity	Duration	Evaluator
1. Attend	Complete the target course as a student; achieve Go on all evaluations	Per course schedule	Course lead instructor
2. Shadow	Observe a certified instructor deliver the full course; take notes on pacing, transitions, and student management	1 course iteration	Supervising instructor
3. Co-Teach	Deliver 2–3 blocks of instruction under observation; receive structured feedback	1 course iteration	Supervising instructor
4. Lead (Observed)	Deliver the full course as lead instructor with an observer present	1 course iteration	Training OIC/NCOIC
5. Certify	Training OIC/NCOIC signs certification memorandum; instructor added to certified roster	Upon completion of Phase 4	Training OIC/NCOIC

NOTE

Phase 2 (Shadow) and Phase 3 (Co-Teach) may occur in the same course iteration if the supervising instructor determines the candidate is ready to deliver blocks during that iteration. Phases may not be skipped.

9-2. Recertification

- Instructors must deliver at least one iteration of their certified course every 12 months to maintain certification.
- If 12 months elapse without delivery, the instructor must complete Phase 3 (Co-Teach) before resuming independent delivery.
- Curriculum updates that change >25% of course content require all certified instructors to complete a delta briefing before next delivery.

9-3. Certification Authority

The USAREUR-AF C2DAO Training OIC is the certification authority for all MSS instructor certifications. Certification memoranda are maintained by the training NCOIC.

9-4. Cross-Certification

Instructors certified for one TM-40 specialist track (G–O) may cross-certify for additional tracks by completing Phases 1 and 3 for the new track. WFF tracks (A–F) require full certification pathway due to domain-specific content.

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Reference: TP 350-70-3, *Faculty and Staff Development Program* (TRADOC). See also `training_management/FACULTY_DEVELOPMENT_PLAN.md` for instructor qualification matrices and evaluator prerequisites.

10. CONTACTS AND ESCALATION

Issue	Contact
Training environment access, provisioning failures	C2DAO (submit access request; lead times vary by capability — see course ENVIRONMENT_SETUP.md)
Enrollment, training records, evaluation forms	Training NCO
Curriculum questions, content errors, TM updates	USAREUR-AF Operational Data Team
Student enrollment exceptions (missing prereqs)	Training OIC; do not make exceptions independently
Classification or OPSEC concerns about course material	Training OIC + unit S2/S6

11. QUICK REFERENCE — COURSE DOCUMENT MAP

For any given course, the complete instructor document set is:

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Document	Location	Purpose
TM	`tm/TM_XX_name/TM_XX_NAME.md`	Student and instructor primary reference
Concepts Guide (TM-40+)	`tm/TM_XX_name/CONCEPTS_GUIDE_TMxx_NAME.md`	Supplemental background reading
Syllabus	`syllabi/SYLLABUS_TMxx.md`	Schedule and learning objectives
Lesson Plan	`training_management/lesson_plans/[outline file]`	Block-by-block delivery guide
PRE Exam	`exercises/exams/EXAM_TMxx_PRE.md`	Day 1 diagnostic (answer key embedded)
POST Exam	`exercises/exams/EXAM_TMxx_POST.md`	Final knowledge check (answer key embedded)
Exercise	`exercises/EX_XX_name/EXERCISE.md`	Practical exercise task list and evaluation criteria
Environment Setup	`exercises/EX_XX_name/ENVIRONMENT_SETUP.md`	Pre-exercise environment configuration
PDF versions	`pdf/`	Print-ready versions of all above

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